

Confidence interval for difference of two means; independent samples, variances unknown but assumed to be equal
Apples example

Background You have the same datasets from the lesson.
Task 1 Calculate the 90% confidence interval
Task 2 Compare the result with the 95% confidence interval from the lesson

NY apples	LA apples
\$ 3.80	\$ 3.02
\$ 3.76	\$ 3.22
\$ 3.87	\$ 3.24
\$ 3.99	\$ 3.02
\$ 4.02	\$ 3.06
\$ 4.25	\$ 3.15
\$ 4.13	\$ 3.81
\$ 3.98	\$ 3.44
\$ 3.99	
\$ 3.62	

	NY	LA
Sample mean	\$ 3.94	\$ 3.25
Sample std	\$ 0.18	\$ 0.27
Sample size	10	8
Pooled variance	0.05	
Pooled std	0.22	
90% CI, t16, 0.05	1.746	
	0.05	

Confidence intervals			
T	CI low	CI high	
90%	\$ 0.51	\$ 0.88	

NY apples	LA apples
\$ 3.80	\$ 3.02
\$ 3.76	\$ 3.22
\$ 3.87	\$ 3.24
\$ 3.99	\$ 3.02
\$ 4.02	\$ 3.06
\$ 4.25	\$ 3.15
\$ 4.13	\$ 3.81
\$ 3.98	\$ 3.44
\$ 3.99	
\$ 3.62	

	NY	LA
Sample mean	\$ 3.94	\$ 3.25
Sample std	\$ 0.18	\$ 0.27
Sample size	10	8
Pooled variance	0.05	
Pooled std	0.22	
95% CI, t16,0.025	2.12	

Confidence intervals			
T	CI low	CI high	
95%	\$ 0.47	\$ 0.92	

From previous exercise confidence interval at 95%

A lower confidence results in a narrower interval (see Margin of error lesson)

# populations	Population variance	Samples	Statistic	Variance	Formula
Two	unknown, assumed equal	independent	t	$s_p^2 = \frac{(n_x - 1)s_x^2 + (n_y - 1)s_y^2}{n_x + n_y - 2}$	$(\bar{x} - \bar{y}) \pm t_{n_x+n_y-2, \alpha/2} \sqrt{\frac{s_p^2}{n_x} + \frac{s_p^2}{n_y}}$